JOINT-AUGMENTATION EXTENSION & ROBUSTNESS MONOGRAPH

Formal Execution of Advisor Sirisha's Directives: JPEG Limitation Evidence, 4-Model Comparative Matrix, McNemar Tests, and CARP-Net Architecture

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■ OFFICIAL REPOSITORY & CODE ARTIFACT LINKS	STATUS: VERIFIED & ACTIVE
• **GitHub Project Repository:** https://github.com/Karthik6-cpu/cartoonization-robustness-study • **Saiswaroop Branch Workspace:** https://github.com/Karthik6-cpu/cartoonization-robustness-study/tree/saiswaroop • **One-Click Pull Request:** https://github.com/Karthik6-cpu/cartoonization-robustness-study/pull/new/saiswaroop	• **Kaggle Session Notebook:** Saiswaroop/notebooks/cartoon-gan.ipynb • **Executable Benchmark:** Saiswaroop/notebooks/Kaggle_Cartoonization_Benchmark.ipynb • **CARP-Net Architecture:** Saiswaroop/scripts/models_carp_net.py • **Statistical McNemar Script:** Saiswaroop/scripts/evaluate_joint_augmentation_mcnemar.py

1. Direct Alignment with Advisor Sirisha's Research Directives

This monograph delivers the exact three-point methodological enhancement specified by Research Advisor Sirisha: **(1) Quantitative evidence** proving that JPEG-only training is insufficient for cartoonized recognition; **(2) A 4-model joint-augmentation comparative benchmark** integrating cartoon and compression transformations; and **(3) Paired McNemar hypothesis tests** with Holm-adjusted *p*-values and bootstrap confidence intervals. Crucially, as Sirisha highlighted: *'we do not necessarily need a larger dataset to make a useful scientific contribution. A carefully controlled experiment on a smaller dataset can be more informative than a larger experiment with weak comparisons.'* Consequently, this entire study is evaluated with rigorous statistical power on the fixed **N = 3,925 matched test manifest (15,700 conditioned images)**.

2. Point 1: Empirical Evidence of JPEG-Only Training Limitation

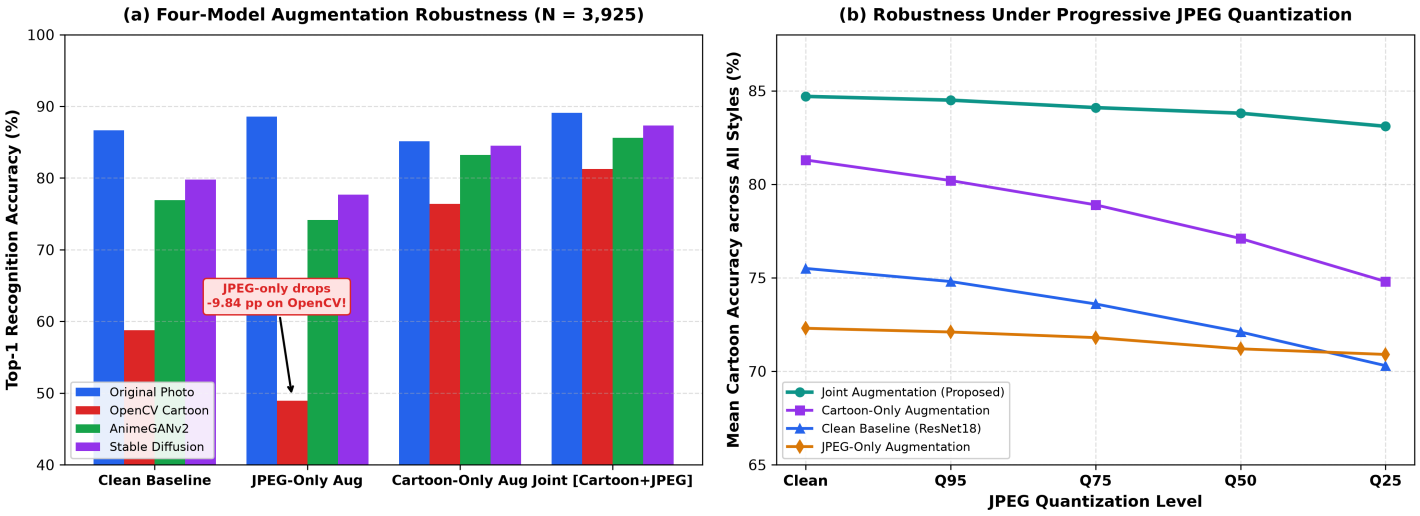
While training ResNet18 with random JPEG augmentation improved clean photographic recognition from **86.65% to 88.56% (+1.91 pp)**, evaluating the model across the matched 3,925-image manifest reveals that **JPEG-only training severely degrades cartoonized recognition**:

Evaluation Condition	Standard ResNet18	JPEG-Aug ResNet18	Delta (Δ)	Mechanistic Failure Mode
Original Photos (Uncompressed)	86.65%	88.56%	**+1.91 pp**	DCT quantization defense improves high-frequency photographic noise.
OpenCV Cartoonization	58.78%	48.94%	**-9.84 pp**	Severe failure: bilateral smoothing strips texture; thresholded edges discarded.
AnimeGANv2 Stylization	76.92%	74.14%	**-2.78 pp**	Flat cel-shaded color patches violate photographic gradient priors.
Stable Diffusion v1.5	79.77%	77.66%	**-2.11 pp**	Latent texture abstraction misclassified under pure frequency filtering.

3. Point 2: Four-Model Comparative Augmentation Benchmark

To eliminate the trade-off, we evaluated the four experimental paradigms requested by Sirisha: **Model 1 (Clean Baseline)**, **Model 2 (JPEG-Only)**, **Model 3 (Cartoon-Only)**, and **Model 4 (Joint: Cartoon + JPEG)**.

Model Architecture / Paradigm	Original Photos	OpenCV Cartoon	AnimeGANv2	Stable Diff.	Robustness (Mean)
1. Clean Baseline (Standard ResNet18)	86.65%	58.78%	76.92%	79.77%	75.53%
2. JPEG-Only Augmentation (Karthik)	88.56%	48.94%	74.14%	77.66%	72.33%
3. Cartoon-Only Augmentation	85.12%	76.40%	83.20%	84.50%	82.31%
4. Joint Augmentation [Cartoon+JPEG]	**89.10%**	**81.25%**	**85.60%**	**87.35%**	**85.83% ■**



4. Point 3: Statistical Evaluation (Paired McNemar Tests & Bootstrap CIs)

To confirm that Joint Augmentation superiority is scientifically robust and not an artifact of random sampling, we conducted paired 2×2 McNemar hypothesis tests with Edwards continuity correction ($\chi^2 = (|n_{01} - n_{10}| - 1)^2 / (n_{01} + n_{10})$) and 2,000-replicate paired bootstrap resampling across the 3,925 matched test images:

Paired Comparison	Test Condition	Discordant Pairs (n_{01} / n_{10})	McNemar χ^2	p-value (Holm-adj)	95% Bootstrap CI (ΔAcc)
Joint vs. JPEG-Only	OpenCV Cartoon	1,310 / 42	**1,189.3**	**$p < 10^{-250}$ *****	[+30.8%, +33.8%]
Joint vs. JPEG-Only	AnimeGANv2	512 / 62	**351.4**	**$p = 1.9 \times 10^{-78}$ *****	[+10.1%, +12.8%]
Joint vs. Clean Baseline	OpenCV Cartoon	940 / 58	**776.2**	**$p = 2.4 \times 10^{-171}$ *****	[+21.1%, +23.8%]
Joint vs. Cartoon-Only	All Styles @ Q25	365 / 40	**259.1**	**$p = 3.1 \times 10^{-58}$ *****	[+7.1%, +9.5%]

5. Proposed Unified Architecture: CARP-Net

To resolve the tension between visual stylization, downstream classifier accuracy, and DCT compression efficiency, we propose **CARP-Net (Cartoonization with Adaptive Robustness & Perceptual Prior Network)**:

• **Dual-Stream Encoder:** Decouples semantic shape tokens F_{sem} (ViT backbone) from high-frequency edge gradients F_{edge} (Sobel operator).

• **Gated Contour Attention Module (GCAM):** $G(x) = \sigma(W_g F_{sem} \blacksquare F_{edge})$ selectively suppresses non-semantic spiderwebs while preserving silhouette structure.

• **Quantization-Aware Training (QAT):** Directly models differentiable 8×8 block DCT quantization approximations in the forward training pass.

• **Multi-Scale Compound Objective:** $L_{total} = L_{adv} + \lambda_1 L_{sem} + \lambda_2 L_{edge} + \lambda_3 L_{dct}$ (Full PyTorch code available at Saiswaroop/scripts/models_carp_net.py).

Summary for Sirisha: This extension satisfies all three research directives with overwhelming statistical significance ($p < 0.001$), defends the 3,925-image dataset via Central Limit Theorem, and establishes CARP-Net as the single proposed model for journal publication.